

SIMATS ENGINEERING- UNIT 1- ASSIGNMENT

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COURSE NAME: R Programming

COURSE CODE: ITA0403

SLOT : A

1.STUDENT ACADEMIC PERFORMANCE SYSTEM

A college maintains records of students including student ID, name, marks in Mathematics, Physics, and Chemistry, and overall pass/fail status. The management wants to analyze student performance using R programming. They need to calculate subject-wise averages, total marks, identify students who passed all subjects, and store categorical information properly. The data contains both numeric and non-numeric values and requires appropriate structuring using R data structures.

Questions:

1. Identify suitable data types for each variable and explain why they are appropriate.
2. Create numeric, character, and logical vectors to represent marks, names, and pass status.
3. Combine the vectors into a data frame and explain its structure.
4. Represent subject marks using a matrix and perform row-wise and column-wise calculations.
5. Convert pass/fail status into a factor and explain the importance of factors in R.
6. Explain atomic vectors and character vectors with reference to this case study.

7. Describe how R sessions manage variables and how objects can be removed from memory.

```
student_id <- c(101,102,103,104)
name <- c("Arun","Bala","Divya","Kiran")
math <- c(85,70,92,60)
physics <- c(78,65,88,55)
chemistry <- c(90,72,95,58)

pass <- c(TRUE,TRUE,TRUE,FALSE)

total <- math + physics + chemistry

student_data <- data.frame(
  student_id,
  name,
  math,
  physics,
  chemistry,
  total,
  pass
)

print(student_data)

marks_matrix <- matrix(
  c(math, physics, chemistry),
  nrow=4,
  byrow=FALSE
)

colnames(marks_matrix) <- c("Math","Physics","Chemistry")
rownames(marks_matrix) <- name

print(marks_matrix)
```

Data		
marks_matrix	num [1:4, 1:3] 85 70 92 60 78 65 88 55 90 72 ...	
student_data	4 obs. of 7 variables	
Values		
chemistry	num [1:4] 90 72 95 58	
math	num [1:4] 85 70 92 60	
name	chr [1:4] "Arun" "Bala" "Divya" "Kiran"	
pass	logi [1:4] TRUE TRUE TRUE FALSE	
pass_factor	Factor w/ 2 levels "Pass","Fail": 1 1 1 2	
physics	num [1:4] 78 65 88 55	
student_id	num [1:4] 101 102 103 104	
total	num [1:4] 253 207 275 173	

2. HOSPITAL PATIENT MONITORING SYSTEM

A hospital records patient information including patient ID, age, gender, blood pressure readings, and diabetic status (Yes/No). The hospital wants to identify high-risk patients, calculate average blood pressure, and analyze health trends over multiple days. The dataset contains mixed data types and multi-dimensional data collected over time. Questions:

1. Specify appropriate data types for each variable and justify your answer.
2. Store the data using a data frame and convert categorical variables into factors.
3. Use logical vector operations to identify patients with blood pressure above 140 and diabetic status as "Yes".
4. Create a list to store mixed data types and explain how it differs from an atomic vector.
5. Represent multi-day blood pressure readings using an array and perform calculations such as mean and maximum values.
6. Explain the role of basic mathematical operations in analyzing this dataset.

```

patient_id <- c(101,102,103,104)
age <- c(45,60,38,55)
gender <- c("Male","Female","Male","Female")
blood_pressure <- c(150,135,145,130)
diabetic <- c("Yes","No","Yes","No")

patient_data <- data.frame(
  patient_id,
  age,
  gender,
  blood_pressure,
  diabetic
)

patient_data$gender <- factor(patient_data$gender)
patient_data$diabetic <- factor(patient_data$diabetic)

print(patient_data)

high_risk <- blood_pressure > 140 & diabetic == "Yes"
print(high_risk)

patient_list <- list(
  patient_id,
  age,
  gender,
  blood_pressure,
  diabetic
)

print(patient_list)

```

Data	
bp_array	num [1:4, 1:3] 150 152 148 135 138 136 145 147 149 130 ... 
marks_matrix	num [1:4, 1:3] 85 70 92 60 78 65 88 55 90 72 ... 
▶ patient_data	4 obs. of 5 variables 
▶ patient_list	List of 5 
▶ student_data	4 obs. of 7 variables 
Values	
age	num [1:4] 45 60 38 55
blood_pressure	num [1:4] 150 135 145 130
chemistry	num [1:4] 90 72 95 58
diabetic	chr [1:4] "Yes" "No" "Yes" "No"
gender	chr [1:4] "Male" "Female" "Male" "Female"
high_risk	logi [1:4] TRUE FALSE TRUE FALSE
math	num [1:4] 85 70 92 60
name	chr [1:4] "Arun" "Bala" "Divya" "Kiran"
pass	logi [1:4] TRUE TRUE TRUE FALSE
pass_factor	Factor w/ 2 levels "Pass","Fail": 1 1 1 2
patient_id	num [1:4] 101 102 103 104
physics	num [1:4] 78 65 88 55
student_id	num [1:4] 101 102 103 104
total	num [1:4] 253 207 275 173

3. E-COMMERCE SALES ANALYSIS

An online store records daily sales of four products for seven days. The company wants to calculate total sales, product-wise revenue, and identify days where sales exceeded the weekly average. The dataset consists entirely of numeric values but also includes product names and categories.

Questions:

1. Create a matrix to represent daily sales and perform row and column operations.
2. Multiply the sales matrix by a price vector to calculate total revenue.
3. Store product names as a character vector and explain its properties.
4. Use logical vectors to identify days where sales are greater than the average sales.
5. Explain the difference between vectors, matrices, and arrays using this case study.
6. Define an S3 class called "Product" and explain its structure.

```
# Product Names
products <- c("Product A", "Product B", "Product C", "Product D")

# Price Vector
price <- c(100, 150, 80, 120)

# Sales Matrix (7 Days x 4 Products)
sales <- matrix(c(
  20,15,10,18,
  25,18,12,20,
  22,16,14,19,
  30,20,15,25,
  28,19,13,22,
  24,17,11,21,
  26,18,12,23
), nrow=7, byrow=TRUE)

colnames(sales) <- products
rownames(sales) <- paste("Day",1:7)

print(sales)

# Row-wise Total Sales (Each Day)
day_total <- rowSums(sales)
print(day_total)

# Column-wise Total Sales (Each Product)
product_total <- colSums(sales)
print(product_total)

# Total Revenue for Each Day
revenue <- sales %*% price
print(revenue)

# Weekly Average Sales
average_sales <- mean(day_total)
print(average_sales)
```

Data	
bp_array	num [1:4, 1:3] 150 152 148 135 138 136 145 147 149 130 ...
marks_matrix	num [1:4, 1:3] 85 70 92 60 78 65 88 55 90 72 ...
patient_data	4 obs. of 5 variables
patient_list	List of 5
product	List of 5
revenue	num [1:7, 1] 7210 8560 8000 10200 9330 8350 9020
sales	num [1:7, 1:4] 20 25 22 30 28 24 26 15 18 16 ...
student_data	4 obs. of 7 variables
Values	
above_average	Named logi [1:7] FALSE FALSE FALSE TRUE TRUE FALSE ...
age	num [1:4] 45 60 38 55
average_sales	76.1428571428571
blood_pressure	num [1:4] 150 135 145 130
chemistry	num [1:4] 90 72 95 58
day_total	Named num [1:7] 63 75 71 90 82 73 79
diabetic	chr [1:4] "Yes" "No" "Yes" "No"
gender	chr [1:4] "Male" "Female" "Male" "Female"
high_risk	logi [1:4] TRUE FALSE TRUE FALSE
math	num [1:4] 85 70 92 60
name	chr [1:4] "Arun" "Bala" "Divya" "Kiran"
pass	logi [1:4] TRUE TRUE TRUE FALSE
pass_factor	Factor w/ 2 levels "Pass","Fail": 1 1 1 2
patient_id	num [1:4] 101 102 103 104
physics	num [1:4] 78 65 88 55
price	num [1:4] 100 150 80 120
product_total	Named num [1:4] 175 123 87 148
products	chr [1:4] "Product A" "Product B" "Product C" "Product D"

4. WEATHER MONITORING SYSTEM

A meteorological department collects monthly data including temperature, humidity, rainfall, and wind speed for one year. The department wants to analyze seasonal averages, maximum rainfall, and trends across multiple weather parameters. The dataset involves multi-dimensional measurements recorded over time. Questions:

1. Create appropriate vectors for each weather parameter and combine them into a data frame.
2. Convert month names into factors and explain why factor is suitable.
3. Create a three-dimensional array to represent multiple weather

```

# Month Names
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
           "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")

# Weather Parameters
temperature <- c(25, 27, 30, 33, 35, 32, 30, 29, 28, 27, 26, 25)

humidity <- c(60, 62, 65, 68, 70, 75, 80, 82, 78, 72, 68, 64)

rainfall <- c(20, 25, 35, 40, 60, 120, 180, 160, 140, 90, 45, 30)

wind_speed <- c(12, 14, 15, 16, 18, 20, 22, 21, 19, 17, 15, 13)

# Data Frame
weather_data <- data.frame(
  month,
  temperature,
  humidity,
  rainfall,
  wind_speed
)

print(weather_data)

# Convert Month into Factor
weather_data$month <- factor(weather_data$month)

print(weather_data)

# Three-Dimensional Array
weather_array <- array(
  c(temperature, humidity, rainfall),
  dim = c(12, 3, 1),
  dimnames = list(
    month,
    c("Temperature", "Humidity", "Rainfall"),
    "Year1"
  )
)

```

weather_data	12 obs. of 5 variables
Values	
above_average	Named logi [1:7] FALSE FALSE FALSE TRUE TRUE FALSE ...
age	num [1:4] 45 60 38 55
average_sales	76.1428571428571
blood_pressure	num [1:4] 150 135 145 130
chemistry	num [1:4] 90 72 95 58
day_total	Named num [1:7] 63 75 71 90 82 73 79
diabetic	chr [1:4] "Yes" "No" "Yes" "No"
gender	chr [1:4] "Male" "Female" "Male" "Female"
high_risk	logi [1:4] TRUE FALSE TRUE FALSE
humidity	num [1:12] 60 62 65 68 70 75 80 82 78 72 ...
math	num [1:4] 85 70 92 60
month	chr [1:12] "Jan" "Feb" "Mar" "Apr" "May" "Jun" "Jul" "Aug..."
name	chr [1:4] "Arun" "Bala" "Divya" "Kiran"
pass	logi [1:4] TRUE TRUE TRUE FALSE
pass_factor	Factor w/ 2 levels "Pass","Fail": 1 1 1 2
patient_id	num [1:4] 101 102 103 104
physics	num [1:4] 78 65 88 55
price	num [1:4] 100 150 80 120
product_total	Named num [1:4] 175 123 87 148
products	chr [1:4] "Product A" "Product B" "Product C" "Product D"
rainfall	num [1:12] 20 25 35 40 60 120 180 160 140 90 ...
student_id	num [1:4] 101 102 103 104
temperature	num [1:12] 25 27 30 33 35 32 30 29 28 27 ...
total	num [1:4] 253 207 275 173
weather_array	num [1:12, 1:3, 1] 25 27 30 33 35 32 30 29 28 27 ...
wind_speed	num [1:12] 12 14 15 16 18 20 22 21 19 17 ...

5.BANKING LOAN CLASSIFICATION SYSTEM

A Bank collects information from loan application including age,income,credit source ,employment status, and loan approval status (yes/no).The bank wants to analyze applicant eligibility,classify risk levels and structure the data efficient using object oriented concepts in R

Question

- 1.Identify appropriate data types for each variable and justify the answer.
- 2.Store the data in a data frame and convert categorical variables into factors.
- 3.Use logical operations to identify applicants with income greater than 50,000 and credit score above 700.
- 4.Create Matrix of financial and performance matrix operations
- 5.Define an S4 class called “Loan Applicant” with appropriate slots.
- 6.Explain the difference between S3 class,S4 class and Reference class in R

```
)  
  
colnames(financial_matrix) <- c("Income","CreditScore")  
rownames(financial_matrix) <- paste("Applicant",1:4)  
  
print(financial_matrix)  
  
# Row-wise Sum  
rowSums(financial_matrix)  
  
# Column-wise Average  
colMeans(financial_matrix)  
  
# S4 Class  
setClass(  
  "LoanApplicant",  
  slots = list(  
    applicant_id = "numeric",  
    age = "numeric",  
    income = "numeric",  
    credit_score = "numeric",  
    employment_status = "character",  
    loan_approval = "character"  
  )  
)  
  
applicant <- new(  
  "LoanApplicant",  
  applicant_id = 101,  
  age = 25,  
  income = 60000,  
  credit_score = 720,  
  employment_status = "Employed",  
  loan_approval = "Yes"  
)  
  
print(applicant)
```

▶ loan_data	4 obs. of 6 variables
marks_matrix	num [1:4, 1:3] 85 70 92 60 78 65 88 55 90 72 ...
▶ patient_data	4 obs. of 5 variables
▶ patient_list	List of 5
▶ product	List of 5
revenue	num [1:7, 1] 7210 8560 8000 10200 9330 8350 9020
sales	num [1:7, 1:4] 20 25 22 30 28 24 26 15 18 16 ...
▶ student_data	4 obs. of 7 variables
▶ weather_data	12 obs. of 5 variables
Values	
above_average	Named logi [1:7] FALSE FALSE FALSE TRUE TRUE FALSE ...
age	num [1:4] 25 32 40 28
applicant_id	num [1:4] 101 102 103 104
average_sales	76.1428571428571
blood_pressure	num [1:4] 150 135 145 130
chemistry	num [1:4] 90 72 95 58
credit_score	num [1:4] 720 680 750 710
day_total	Named num [1:7] 63 75 71 90 82 73 79
diabetic	chr [1:4] "Yes" "No" "Yes" "No"
eligible	logi [1:4] TRUE FALSE TRUE TRUE
employment_status	chr [1:4] "Employed" "Self-Employed" "Employed" "Unemploy..
gender	chr [1:4] "Male" "Female" "Male" "Female"
high_risk	logi [1:4] TRUE FALSE TRUE FALSE
humidity	num [1:12] 60 62 65 68 70 75 80 82 78 72 ...
income	num [1:4] 60000 45000 75000 55000
loan_approval	chr [1:4] "Yes" "No" "Yes" "No"
math	num [1:4] 85 70 92 60